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RESEARCH ARTICLE

Heart Disease Prediction Using a Hybrid Feature Selection and Ensemble Learning Approach

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ABSTRACT Heart diseases have become the leading cause of death globally, highlighting the urgent need for robust diagnostic and treatment methods. This study leverages the UCI heart disease dataset to assess the effectiveness of various Machine Learning models in predicting heart diseases. This paper proposed an advanced prediction method that combines feature selection using a hybrid of Genetic Algorithm (GA) and Cuckoo Search Optimization (CSO) with a majority voting ensemble of Convolutional Neural Network and Random Forest. This approach also integrated GA for hyperparameter tuning, enhancing predictive accuracy. Comprehensive preprocessing techniques, including handling missing values, outlier detection, and normalization, were employed to ensure data quality. Using the proposed approach, 95% accuracy, 95.65% precision, 91.7% recall, 93.61% F1-score, 97.22% specificity, and 95.02% ROC AUC has been achieved. Our results demonstrate that the proposed method outperforms the existing models.

INDEX TERMS Cardiovascular disease, nature-inspired algorithm, feature selection, genetic algorithm, machine learning, optimization, ensemble learning.

I. INTRODUCTION

Cardiovascular diseases are one of the leading causes of death worldwide [1]. Cardiac disease comprises a variety of problems that affect functioning of the muscles and heart rate. Common heart diseases comprise heart rhythm problems, congenital heart defects, and coronary artery disease. Many lives can be saved via accurate diagnosis [29]. Several risk factors associated with heart disease include physiological factors like poor eating habits, high cholesterol, and hypertension, while behavioral factors include excessive smoking, and alcohol intake [3]. Since heart disease has a high mortality rate, the development of an effective diagnostic method for identifying diseased patients is a huge challenge [21]. Traditional clinical methods rely on static factors like physical examinations and patient history including age, gender, smoking history, and family background.

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While these methods provide valuable information, they often rely heavily on the clinician's expertise and experience, which can lead to variability in diagnosis and treatment decisions. Clinical approaches are static and cannot catch up with increasing medical knowledge. Furthermore, these diagnostic procedures can be time-consuming and sometimes invasive, highlighting the need for more efficient and accurate diagnostic tools [4]. Over the past decade, concepts like artificial intelligence, semantic computing, machine learning, Internet of Things, and deep learning have been making their way into the healthcare sector [2,38]. Machine learning (ML) is a field in which models are trained using historical data and then predictions are made using these trained models. ML is non-invasive compared to traditional clinical methods and can handle vast amounts of data subathra. Moreover, ML can uncover complex hidden patterns and insights that might not be possible through conventional methods. A notable disadvantage of ML is its limitations in dealing with non-linear relationships in data. This is where Deep Learning

(DL) steps in. Deep learning, a subset of machine learning, uses neural networks to extract intricate patterns, thereby mitigating the drawbacks of classical ML. DL simulates the neural architecture of the human brain, allowing computers to absorb and comprehend data similarly. DL can easily handle non-linear relationships between data attributes and manage intricate data [21]. The adoption of ML and DL in healthcare offers several advantages. These models can process vast amounts of data quickly, providing real-time insights and predictions [2]. They can also improve the accuracy of diagnoses and treatment plans, reduce the burden on healthcare professionals, and ultimately enhance patient care. However, the model's effectiveness relies on the quality of input data. Furthermore, previous studies raised a number of challenges such as difficulty in managing data imbalance, lack of robustness, and high data dimensionality. To address these shortcomings, our study suggested a novel method in predicting cardiac diseases. The main contributions of this paper are:

- To propose a novel hybrid feature selection method that integrates Genetic Algorithm (GA) with Cuckoo Search Optimization (CSO), combining global and local search strengths for improved feature relevance.
- To design a custom ensemble classifier that combines Random Forest and Convolutional Neural Network (CNN) using majority voting, enhancing both accuracy and robustness.
- To apply Genetic Algorithm-based hyperparameter tuning across the model pipeline, improving performance consistency.

The proposed hybrid algorithm is designed to leverage GA's strong global search capabilities and CSO's efficient local exploitation via Lévy flight. This integration is implemented using a carefully structured optimization flow and is shown to outperform standalone methods for feature selection. The selected features then feed into the proposed ensemble classification framework that takes advantage of the interpretability and robustness of RF with the deep feature extraction capabilities of CNN. Finally, GA-based hyperparameter tuning is performed not only for individual models but also in conjunction with the proposed hybrid feature selector, contributing to overall model robustness and optimization. The proposed method is extensively validated against state-of-the-art ML and DL models using the UCI Heart Disease dataset. The superior performance across accuracy, recall, F1-score, and specificity, confirmed the effectiveness and generalizability of the proposed approach. Moreover, a comprehensive complexity analysis of all major components of the proposed pipeline, including feature selection, classification, and ensemble decision-making, offers insight into the model's scalability for real-world deployment.

The remaining part of this paper is arranged as follows: Section II presents the existing literature review. Section III describes the proposed methodology. The experimental setup used in this paper is displayed in Section IV. Section V

validates the findings and results, followed by a conclusion in Section VI.

II. RELATED WORK

This section thoroughly summarizes existing machine learning and deep learning models employed for heart disease prediction.

A. MACHINE LEARNING TECHNIQUES

Subathra et al. [3] proposed an efficient methodology named, Bolstered Swarm Integrated Ensemble Learning (BSEL), where they utilized Linear Interpolation Normalization (LIN) for preprocessing. Feature selection was performed using Bolstered-up Beetle Swarm Optimization (BBSO). Weighted Ensemble Classification (WEC) was employed for predicting test data. Additionally, they used Red Colobus Monkey Optimization (RCMo) for parameter tuning. Zhang et al. [23] used SMOTE to balance data and then combined 4 ML models: Logistic regression (LR), artificial neural network (ANN), random forest (RF) and gradient boosting (GB). Itoo and Garg [4] developed a Stacking CV Classifier, constituting three ML models: KNN, NB, and LR. Whereas, Sulthana et al. [5] assessed the efficiency of NB, DT, and NB with k-means clustering. Gaikwad et al. [6] used UCI dataset to evaluate the performance metrics of various ML models, such as SVM, DT, RF, GB, and Logistic Regression (LR) while Jadhav et al. [7] assessed performance of machine learning models on. A clustered GA for heart disease prediction was found by Vijaya et al. [9]. Islam et al. [10] employed PCA for dimensionality reduction and then applied hybrid GA via K means to achieve final clustering. In order to deal with class imbalance, Abdellatif et al. [11] used SMOTE. They then utilized Hyperband Algorithm for final prediction. Nanda Kumar and Narayan [17] proposed a different approach to detect heart disease. They used a hamming distance-based feature selection method and a combination of cuckoo search algorithm with a deep belief network for finding correct predictions and got an accuracy ranging between 89-91% for different datasets.

B. DEEP LEARNING TECHNIQUES

Dileep et al. [2] introduced Cluster-based Bi-LSTM, and compared its approach with traditional models. Meanwhile, Chamundeshwari et al. [12] optimized CNN using cat swarm optimization technique. They utilized the Field II program for removing speckle noise from ECG images along with the hybrid pattern extraction technique. Jain et al. [13] evolved the motivation for optimization. They employed the Levy Flight CNN (LV-CNN) technique along with the Sunflower Optimization Algorithm. This approach yielded an accuracy of 95.74%. Hussain et al. [14] suggested a 1D CNN approach along with the usage of the dropout method. A feature selection method using genetic algorithms in combination with an ensemble deep neural network framework was improved using the Adam optimizer [15].

Liu et al. [19] discovered the Random Horizontal Flip classification method. They used kernel-based pruning for ECG classification. Karri et al. [20] employed a different strategy for arrhythmia classification. They detected P and T waves with the help of Delta-sigma modulation and found R peaks using Discrete wavelet transform. After pre-processing and feature extraction, they used LSTM (RNN) to classify different arrhythmias. This approach resulted in a remarkable accuracy of 99.64%. Tiwari et al. [21] combined CNN and CNN-LSTM to make an ensemble framework. CNN layers were used for extracting important features which are then fed into the LSTM model. Korurek et al. [22] proposed Ant Colony Optimization (ACO) for arrhythmia clustering. They used a low-pass filter to remove noise and a median filter for baseline estimation. Khalil et al. [24] extracted features using Stationary Wavelet Transform. These features were then fed into a 1D-CNN model and an accuracy of 99.57% was achieved by using a 10-fold cross-validation technique. Begum et al. [25] compared 2 models – 1D CNN-LSTM model and 1D CNN. In [27], the authors added some random noise in the data and oversampled it and later used CNN for predicting heart disease. Table 1 shows a summary of work done by authors in predicting heart disease.

C. INSIGHTS AND CHALLENGES

From above literature this can be concluded that both techniques be it machine learning or deep learning have demonstrated their potential in solving the problem of heart disease prediction. Here, ML models often combined with feature selection and data balancing, provide more interpretability and simplicity, while DL models especially CNN and LSTM, deliver significantly better performance on ECG signals. Conclusively, hybrid models and ensemble approaches often yield better results compared to individual methods.

However, several challenges remain prevalent in this domain:

- **High dimensionality:** Large feature spaces lead to overfitting if actual responsible features are not identified.
- **Class imbalance:** It causes majority class bias due to skewness of the data.
- **Interpretability:** Deep learning models act as black boxes which limits clinical trust.
- **Overfitting:** Small datasets combined with complex models increase the risk of poor generalization.
- **Optimization difficulties:** Tuning hyperparameters effectively is computationally expensive and challenging.
- **Computational demands:** Deep learning models often requires complex hardware and require more training time.

In summary, existing techniques show promising results. However, it also reveals that no single approach fully addresses the complexity of real-world clinical data. Therefore, addressing these challenges is essential by developing a model that combines robust feature selection, effective

model optimization, and ensemble learning to achieve high accuracy and generalizability. Motivated by these findings, a hybrid feature selection method that combines Genetic Algorithm (GA) and Cuckoo Search Optimization (CSO) along with an ensemble of Convolutional Neural Network and Random Forest has been proposed. GA is renowned for its strong search capabilities and exploration of a vast solution space, while CSO is superior in avoiding local optima and achieving fast convergence. By combining these techniques, the strengths of both GA's exploration and CSO's exploitation are maximized, resulting in more efficient feature selection and improved performance of the heart disease prediction model.

III. PREPROCESSING

Before applying the proposed algorithm, it is crucial to preprocess the data to ensure high-quality inputs for the model. Preprocessing helps remove noise, correct inconsistencies, and standardize the data, which in turn improves the performance and reliability of the prediction model. The main preprocessing steps in this study include outlier detection and removal using the Inter-Quartile Range (IQR) method, followed by data normalization to bring all features onto a comparable scale. These steps are detailed below.

A. INTER-QUARTILE RANGE (IQR)

Outliers are data instances that significantly deviate from the expected range of values, causing bias in analysis and leading to wrong conclusions. In this study, outliers are identified and eliminated using the Inter-Quartile Range (IQR). The IQR represents statistical dispersion and is given by the formula $IQR = Q3 - Q1$, where $Q1$ is the first quartile and $Q3$ is the third quartile. The upper bound is calculated as $Q3 + 1.5 \cdot IQR$, while the lower bound as $Q1 - 1.5 \cdot IQR$. Data points that lie outside of these bounds are considered outliers and are removed [8]. This step ensures that the remaining data represents underlying patterns more closely, thereby increasing the reliability of our analysis.

B. NORMALIZATION

After removing all the outliers, normalization is employed to standardize the data and prepare it for subsequent analysis. Normalizing the data ensures that the model doesn't get biased, which leads to enhanced accuracy. We applied two methods: Linear Interpolation Normalization and Min-Max Normalization. Among these, Min-Max Normalization yielded better results, so it was used for further research work.

Min-Max Normalization transforms data into a uniform scale. This technique balances the influence of all the attributes of data by preventing any single feature from overpowering others. Normalization is done using formula:

$$X' = \frac{X - \min(X)}{\max(X) - \min(X)} \quad (1)$$

where X is the original value and X' is the normalized value.

TABLE 1. Summary of existing literature on heart disease prediction using ML techniques.

Authors	Datasets	Models used	Performance Metrics	Data Type
Dileep et al. (2022)	UCI heart disease dataset and real time dataset	Cluster-based Bi-LSTM	Accuracy, Sensitivity, Recall, F1 score	Textual
Yang and Guan (2022)	Private pathological data	Smote XGBoost	Accuracy, Precision, Recall, F1-Score, AUC	Textual
Subathra et al. (2024)	Cleveland, Statlog, and comprehensive datasets	Bolstered Swarm Integrated Ensemble Learning	Accuracy, Precision, Specificity, Sensitivity, F1-Score	Textual
Yang et al. (2023)	Framingham Heart Institute (FHS)	Light GBM, OPTUNA	Sensitivity, Recall, F-Score, Accuracy, Specificity, Precision	Textual
Ito and Garg (2022)	Cleveland Clinic Foundation (CVF)	Ensemble model: StackingCV-Classifer	Prediction, Accuracy,	Textual
Kapila et al. (2023)	Cleveland, CVD, HD dataset	Quine McCluskey Binary Classifier (QMBC) with Chi-square, Anova and PCA	Accuracy, Recall, F1-Score, Precision, Specificity,	Textual
Vijaya (2023)	UCI heart disease dataset	Clustered GA	Accuracy, Precision, Recall, Error Rate, F-Score,	Textual
Islam et al. (2020)	UCI heart disease dataset	Hybrid GA with k means + PCA	Accuracy	Textual
Sharma and Parmar (2023)	Heart Disease UCI dataset	Optimized DNN using Talos	Accuracy	Textual
Hussain et al. (2021)	Cleveland dataset	1-D CNN	Accuracy, Precision, Recall, F1-Score,	Textual
Zhang et al. (2014)	MIT BIH arrhythmia	SVM + one-versus-one (OvO)	Sensitivity	ECG
Tiwari et al. (2023)	MIT BIH AR DB	CNN + CNN LSTM	Precision, Sensitivity, F-Score, Accuracy	ECG
Karri et al. (2023)	MIT-BIH dataset, QT dataset	DSM, DWT, LSTM	Accuracy, Positive Predictivity, Sensitivity, F1-Score	ECG
Begum et al. (2023)	MIT BIH arrhythmia	1D CNN, 1D CNN + LSTM	Accuracy, Precision, F1-Score	ECG
Chen et al. (2023)	MIT BIH arrhythmia	cross-modal multiscale multi-instance learning approach	F1 Score, AUC, Recall	ECG
Zhou et al. (2024)	MIT-BIH-AR	FCBA+CNN	Accuracy, Specificity, Sensitivity	ECG
Liu et al. (2024)	MIT-BIH	Random Horizontal Flip (RHF)-based classification	Accuracy, Precision, Recall, Specificity	ECG

IV. PROPOSED WORK

This section presents the suggested nature-inspired algorithm where the data need to be preprocessed according to techniques mentioned in preprocessing section. Following which feature selection and ensemble classification models are applied for prediction. Common challenges faced by researchers in previous literature include lack of robustness, difficulty in handling data imbalance, and high data dimensionality. To address these issues, the proposed approach aims to implement an enhanced framework for heart disease prediction, as illustrated in Figure 1.

The main components of the proposed model are:

- A hybrid feature selection method combining Genetic Algorithm and Cuckoo Search Algorithm.
- Ensemble classification for improved prediction accuracy.
- Genetic Algorithm-based hyperparameter tuning for optimizing model performance.

A. FEATURE SELECTION

To identify the most optimal features for the target variable and enhance the performance of the model [39,8], a feature selection method is employed. Three nature-inspired feature selection algorithms were utilized: Genetic Algorithm (GA),

Cuckoo Search Algorithm (CSA), and Flower Pollination Algorithm (FPA). Among these, a hybrid approach combining GA and CSA [28] yielded the best results, and it was subsequently used for further research.

1) GENETIC ALGORITHM (GA)

The genetic algorithm is based on the principle of natural selection and is widely used in addressing machine learning and optimization problems [16]. The major objective is to refine feature subsets iteratively to reach a high-quality solution close to the optimal solution without knowing the search space [22,36]. This algorithm works on the fundamental concept of ‘survival of the fittest’. The steps involved in GA are:

- 1) Initialization: Initially, a population of individuals, known as chromosomes, is generated randomly.
- 2) Selection: During each iteration of generations, an individual's fitness is calculated using a predefined fitness function.
- 3) Crossover: Crossover helps in the mating between the above-selected individuals [16]. Genetic information between randomly selected crossover sites is exchanged to produce new offspring that comprises traits from both parents.

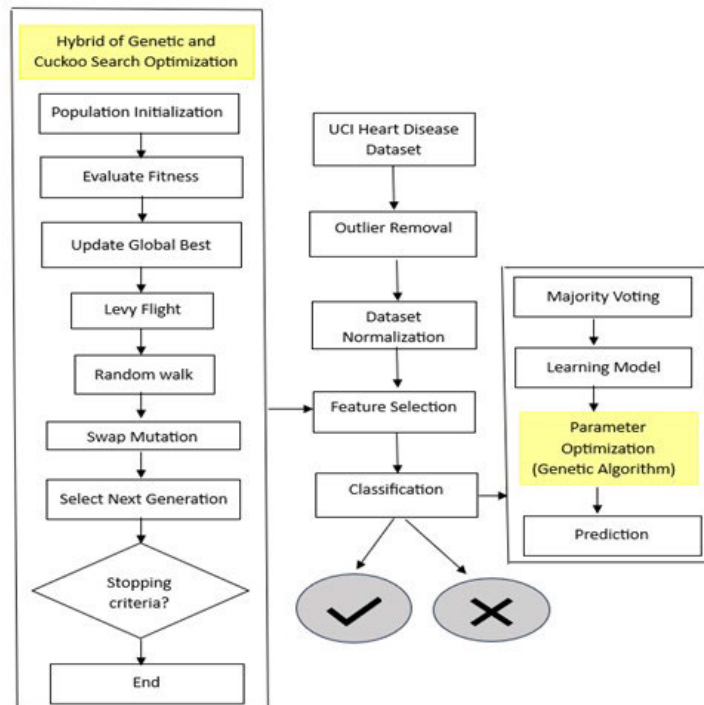


FIGURE 1. Flowchart of the proposed model.

- 4) Mutation: Mutation introduces random genes in the genetic makeup of individuals, ensuring population diversity and avoiding premature convergence.
- 5) Replacement: The next generation is formed by replacing old population with new one.
- 6) Termination: With each successive generation, the solution approaches its optimal state. The algorithm terminates upon attaining a high-quality solution.

2) CUCKOO SEARCH ALGORITHM (CSA)

CSA is a nature-inspired algorithm widely used in domains like speech recognition, cloud computing, and software testing [17]. It works similar to parasitic behavior of certain birds (cuckoos) that lay eggs in the nests of other species (hosts). The host bird can choose to destroy the foreign egg or build a new nest. Levy flight, key mechanism in CSA, enables the algorithm to make long jumps across the search space, thus exploring diverse regions. The steps involved in CSA are:

- 1) Initialization: Initialize a fixed number of nests (n) within the solution space [18].
- 2) Evaluating Fitness: Fitness is computed for each nest to identify the best solutions. The fitness of a newly laid cuckoo egg, representing a potential solution, is compared with that of the host egg within the nest.
- 3) Levy flight: Nests are replaced with new ones based on the levy flight mechanism, fostering diversification.

- 4) Replacement: If the fitness of cuckoo's egg is higher than existing eggs, the old nest is replaced with new solutions.
- 5) Termination: The process terminates after reaching a predefined stopping criterion.

3) HYBRID APPROACH (GA+CSA)

A hybrid strategy combining GA and CSA yielded the best results among the feature-selection algorithms used. This hybrid approach enhances feature selection by leveraging the strengths of both algorithms. CSA introduces diversity through its cuckoo-inspired search behavior and Levy flight mechanism, while GA offers a strong mechanism for exploring the search space through natural selection. Specifically, levy flight mechanism is given by formula [28]:

$$x_i(t+1) = x_i(t) + \alpha_i(t+1) \cdot \text{Levy}(s, \lambda) \cdot (x_i(t) - x_{\text{gbest}}) \quad (2)$$

Additionally, if the randomly generated solution exceeds threshold, the solution is updated using a random walk given by formula [28]:

$$x_i(t+1) = x_i(t) + \text{rand} \cdot (x_k(t) - x_j(t)) \quad (3)$$

If the global best solution does not improve, swap mutation is applied using [28]:

$$x_i(t+1) = x_i(t) + \chi\mu \quad (4)$$

By integrating these two algorithms, the hybrid approach effectively identifies the most relevant features, resulting in better model performance. The steps involved in the hybrid approach are shown in Algorithm 1.

Algorithm 1 Hybrid GA+CSO Algorithm

1. Generate the initial population randomly
 2. for $t = 1, 2, \dots, \text{MaxGenerations}$ do
 3. Calculate fitness for each individual
 4. for $i = 1, 2, \dots, \text{PopSize}$ do
 5. Create solution $x_i(t+1)$ based on Lévy flight
 6. end for
 7. Update global best x_{gbest}
 8. for $i = 1, 2, \dots, \text{PopSize}$ do
 9. if $\text{rand} > \text{threshold}$ then
 10. Update $x_i(t+1)$ using random walk
 11. end if
 12. Repair the solution
 13. Calculate fitness and update the solution
 14. end for
 15. Update global best x_{gbest}
 16. if x_{gbest} does not improve for δ attempts then
 17. Apply swap mutation
 18. end if
 19. Evaluate and update solutions
 20. end for
 21. Store the final solution x_{gbest}
-

Figure 2 provides the steps involved in the hybrid approach of GA+CSA. This hybrid feature selection approach identifies the best features by thoroughly exploring the search space. Table 2 shows the parameters of algorithms used in the hybrid feature selection method. This approach enhanced the predictivity of our model and was used for further research.

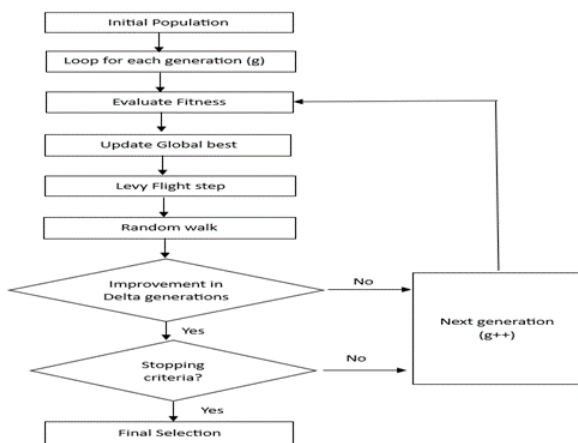


FIGURE 2. Flowchart of GA+CSA.

B. ENSEMBLE BASED CLASSIFICATION

In this step, the classifier is trained with the features selected in the previous step to predict heart disease. We tried various machine learning models like RF, LR, XGB, NB, deep

learning models like CNN, RNN, GRU, Bi-LSTM, and ensemble approaches like RF+XGB+NB, XG+RF+CNN, and many more. Out of these, RF+CNN+Majority voting gave the best results. This proposed model leverages the advantages of both RF and CNN to form a comprehensive predictive framework. The complete details of the proposed model are provided below.

1) RANDOM FOREST (RF)

Random forest is an ensemble learning approach that utilizes multiple decision trees to aggregate their outputs using majority voting to enhance the model's performance. A key concept used in Random Forest is Bootstrap aggregation (Bagging), where a decision tree is fed with bootstrapped data (randomly selected subsets with replacement). This technique reduces the chances of overfitting as it introduces randomness among individual trees. In our study, the RF model is configured with specific parameters optimized through the use of a Genetic Algorithm: a maximum depth of 43 and 194 estimators. These parameters ensure consistent performance by creating a balance between generalization and model complexity.

2) CONVOLUTIONAL NEURAL NETWORK (CNN)

It is a deep learning model used in object detection by extracting intricate patterns and spatial dependencies [26]. In this study, the CNN architecture begins with a 1D convolutional layer having 34 filters, each scanning input attribute to find local patterns. After this layer, Rectified Linear Unit (ReLU) activation is applied to capture complex relationships within data. Subsequently, a max pooling layer with a pool size of 2 downsamples the data. Two fully connected layers further enhance this architecture. There are 50 units with ReLU activation in the first dense layer and one unit with a sigmoid activation function for final binary prediction in the last dense output layer. In the majority voting mechanism, predictions from both models are combined, and the final output is determined based on the majority decision. This methodology improves the overall accuracy and robustness by combining the strengths of both RF and CNN models.

C. PARAMETER TUNING

To enhance the performance of our model, a Genetic Algorithm is used for hyperparameter tuning. Table 2 depicts the parameters of the genetic algorithm and GA+CSO used for hyperparameter tuning of model while Table 3 shows hyperparameters of the model as chosen by the GA. The final RF and CNN models were trained using these selected parameters and the final ensemble was created using majority voting procedure.

V. EXPERIMENTAL SETUP

The setup includes several components that are required to carry out a thorough examination of the proposed algorithms for the prediction of heart disease. The Google Compute

TABLE 2. Parametric study of GA and GA+CSA hybrid as feature selection.

GA	
Hyperparameters	Value
Population size	10
Crossover Probability	0.9
Number of generations	20
Mutation Probability	0.05
GA+CSO	
Hyperparameters	Value
Number of generations	100
Population size	50
Mutation Probability	0.05
Random walk probability	0.25
Crossover Probability	0.9
Delta (Number of generations to check)	5
Levy flight lambda	1.5

TABLE 3. Hyperparameters of the models chosen by GA.

Model	Hyperparameters	Value
RF	Number of estimators	190
RF	Maximum depth	43
CNN	Number of filters	34
CNN	Kernel size	2

TABLE 4. Dataset values.

Attribute Name	Value Range
Age	29-79
Sex	0,1
CP	1-4
Trestbps	94-200
Chol	126-564
Fbs	0,1
Restecg	0-2
Thalach	71-202
Exang	0,1
OldPeak	1-3
Slope	1-3
Ca	0-3
Thal	3,6,7
Num	0,1

Engine handled the entire heart disease prediction approach, which was conducted using Python 3 on Google Colab. The dataset and evaluation metrics utilized are described in this section.

A. DATASET DESCRIPTION

In this paper, the UCI heart disease dataset has been used. This dataset has 14 attributes and 303 instances [3,16]. Table 4 shows the attributes and the corresponding values of the data.

To summarize this dataset, we can conclude that patients aged 29-79 were included, where females were encoded as 0, while males as 1. There are four types of chest pain: Typical Angina (Type 1), Atypical Angina (Type 2); non-Anginal pain (Type 3); and Asymptomatic (Type 4). Attribute

‘Trestbps’ represents Blood pressure and ‘Chol’ represents Cholesterol level. The feature ‘Fbs’ (Fasting blood sugar) has two values 0 or 1. ‘Restecg’ indicates electrocardiographic results. ‘Thalach’ represents the maximum achieved heart rate, while Exercise-induced Angina, ‘Exang’, has a value of 0 or 1 depending on pain. ‘OldPeak’ shows ST depression induced by exercise [3]. The ‘Slope’ feature described slope of the peak of the ST segment and has values 3,6 or 7. ‘Thal’ represents 3 types of defects namely normal, fixed and reversible. Target variable, ‘Num’, indicates if patient has heart disease (1) or not (0).

B. PERFORMANCE METRICS

Precise performance measures are essential to evaluate the performance of the proposed method. Following are the evaluation metrics used in this research:

Accuracy: Gives the ratio of correctly predicted instances to the total number of instances.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

Specificity: Gives the proportion of real negatives that are appropriately identified.

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (6)$$

Recall: Gives the percentage of real positives that are appropriately identified.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (7)$$

Precision: Gives the proportion of positively identified points that are actually correct.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (8)$$

F1 Score: Gives the harmonic mean of recall and precision, providing a balance between the two.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (9)$$

where TP represents true positives, FP represents false positives, TN represents true negatives and FN represents false negatives.

VI. COMPLEXITY ANALYSIS

This section is dedicated to the step by step complexity analysis of the proposed work.

A. OUTLIER DETECTION COMPLEXITY

Inter-Quartile Range (IQR) implemented for outlier detection requires sorting of dataset to calculate the desired quartiles. Sorting a dataset with sample size n has a time complexity of $O(n \log n)$, making the overall complexity of outlier detection $O(n \log n)$.

B. NORMALIZATION COMPLEXITY

Normalization involves iterating through every individual sample of dataset once to compute the minimum and maximum values for each feature which is linear operation and have time complexity of $O(n)$, where n is the number of samples in the dataset.

C. FEATURE SELECTION COMPLEXITY

The complexity of the feature selection process depends on the specific algorithm used as it be Genetic Algorithm or Cuckoo Search Algorithm

1) GENETIC ALGORITHM (GA) COMPLEXITY

It operates in generations where in each generation, the population is evaluated, selected, and evolved. The complexity of each iteration depends on evaluation of the fitness function, which involves computing the performance of each individual. For a population size P and G generations, the complexity of GA is $O(P \cdot G)$, where each individual evaluation takes $O(n)$ time, where n is the number of features in the dataset.

2) CUCKOO SEARCH ALGORITHM (CSA) COMPLEXITY

The CSA performs a number of iterations, each involving the evaluation of fitness and the application of the Levy flight mechanism to update the nest solutions. For n nests and T iterations, the time complexity of the CSA is $O(n \cdot T)$, where the fitness evaluation takes $O(n)$ time.

3) HYBRID GA + CSA COMPLEXITY

In the hybrid GA + CSA approach, both GA and CSA are applied to improve feature selection. Assuming the GA runs for G generations with population P and CSA runs for I iterations with n nests, the overall time complexity of the hybrid approach is given by:

$$O(P \cdot G + n \cdot I)$$

The complexity of the hybrid approach is dominated by the larger of the two components.

D. ENSEMBLE CLASSIFICATION COMPLEXITY

The ensemble classification models from multiple models. The time complexity of the ensemble method depends on the number of classifiers C and the complexity of each base classifier. If each classifier has complexity $O(n)$, the overall time complexity of the ensemble is $O(C \cdot n)$.

E. OVERALL COMPLEXITY

Combining the complexities of all components, the overall time complexity of the proposed approach is determined after considering each step, most computationally expensive components are the feature selection algorithms and ensemble classification. Therefore, the overall complexity of the proposed approach is approximately:

$$O(n \log n + P \cdot G + n \cdot T + C \cdot n)$$

where n is the number of data instances, P is the population size for GA, G is the number of generations for GA, T is the number of iterations for CSA, and C is the number of classifiers in the ensemble model.

F. SPACE COMPLEXITY

The space complexity is determined by the storage required for the dataset, the population of individuals in GA, the nests in CSA, and the ensemble classifiers. The space complexity for storing the dataset is $O(n \cdot f)$, where f is the number of features. The space complexity for GA is $O(P \cdot f)$, and for CSA, it is $O(n \cdot f)$. The space complexity for the ensemble model is $O(C \cdot f)$. Therefore, the overall space complexity is:

$$O(n \cdot f + P \cdot f + n \cdot f + C \cdot f)$$

where n is the number of data instances, f is the number of features, P is the population size for GA, and C is the number of classifiers in the ensemble.

G. SCALABILITY ANALYSIS

The time and space complexity analysis above confirm that the proposed hybrid framework is scalable to larger and more diverse datasets. The feature selection process, based on the integration of Genetic Algorithm (GA) and Cuckoo Search Optimization (CSO), remains computationally efficient due to its parallelizable and population-based structure. Additionally, the ensemble of CNN and Random Forest supports modular and distributed training, making the overall framework well-suited for deployment on scalable infrastructures such as cloud-based platforms. With a computational load that grows linearly or sub-linearly with data size, the proposed approach is practical for real-time or high-volume clinical applications.

VII. RESULTS AND DISCUSSIONS

This section confirms the effectiveness of the proposed hybrid feature selection-based ensemble methodology to detect heart diseases. A summary of the data used is provided in the succeeding sections. For our research, the dataset was divided into two subsets: a training set (80%) and a testing set (20%). Figure 3 demonstrates the convergence comparison of the proposed approach and existing feature selection techniques. The convergence plot shows how effectively the optimizer provides optimal solutions with varying numbers of iterations. From the plot, it is evident that GA+CSO demonstrates superior performance, consistently showing faster convergence compared to other optimizers. Moreover, the stability and minimal fluctuations in GA+CSO make it the most reliable choice.

Based on Table 5, the proposed model performs best compared to existing linear ML and ensemble models in terms of all performance metrics. Among linear models, following the proposed model, LR and linear SVC achieve the same accuracy of 90%, while KNN is the least-performing model. In case of ensemble models, RF is the best-performing model whereas DT performs the least.

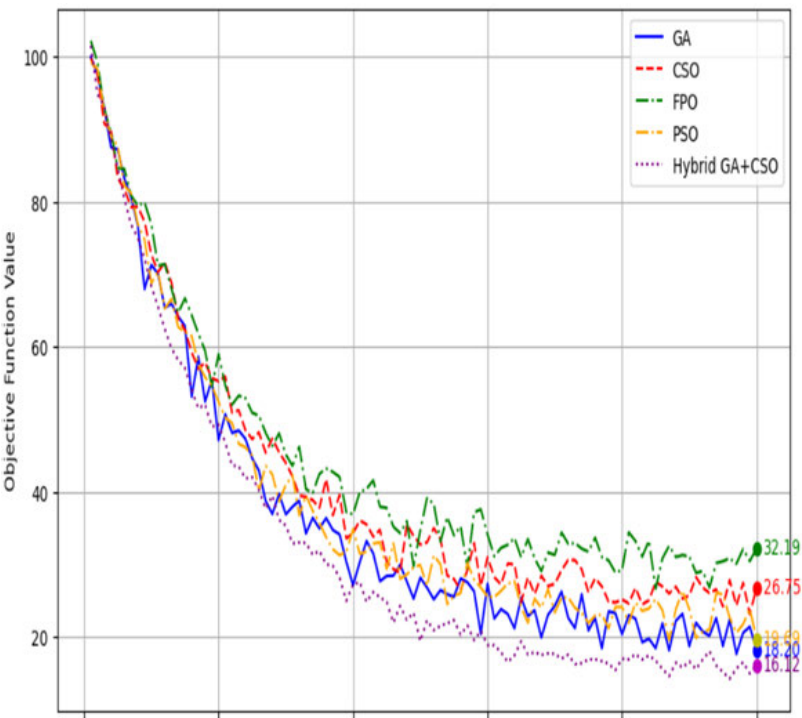


FIGURE 3. Convergence analysis.

TABLE 5. Performance comparison across models.

Comparison among Linear ML Models						
Models	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	Specificity (%)	ROC AUC (%)
Logistic Regression	90.00	84.62	91.67	88.00	88.89	94.91
Linear SVC	90.00	87.50	87.50	87.50	91.67	94.56
Naïve Bayes	86.67	83.33	83.33	83.33	88.89	95.49
KNN	83.33	81.82	75.00	78.26	88.89	94.21
Proposed	95.00	95.65	91.67	93.61	97.22	95.02

Comparison among Ensemble Models						
Models	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	Specificity (%)	ROC AUC (%)
Random Forest	88.33	84.00	87.50	85.71	88.89	94.33
Gradient Boosting	80.00	71.43	83.33	76.92	77.78	89.93
AdaBoost Classifier	83.33	76.92	83.33	80.00	83.33	93.63
Decision Tree	78.33	70.37	79.17	74.51	77.78	78.47
Proposed	95.00	95.65	91.67	93.61	97.22	95.02

TABLE 6. Comparison between different tuning methods.

Models	Genetic Algorithm		Cuckoo Search Algorithm	
	Accuracy	Recall	Accuracy	Recall
RF	84.38	87.50	93.75	90.62
LR	86.00	84.00	79.60	78.12
NB	77.00	69.00	79.50	75.00
SVM	88.00	88.00	79.60	75.00
RF+CNN (Proposed)	95.00	91.70	89.06	81.25

Using the hybrid GA+CSA as a feature selection method, Table 6 compares the accuracies and recall of the proposed model with existing models, with parameter tuning

performed by GA and CSA. It is evident from Table 6 that the proposed model achieves the highest accuracy (95%) and recall (91.7%) with GA tuning compared to other models. Following the proposed model, SVM is the second-best performing model with GA tuning, achieving both accuracy and recall of 88%. RF has the highest accuracy of 93.75% under GA, while NB performs poorly with 77% accuracy and 69% recall. Using CSA tuning, RF emerges as the best performer with 93.75% accuracy and 90.62% recall, followed by proposed RF+CNN with majority voting having an accuracy of 89.06% and recall of 81.25%. LR, NB, and SVM exhibit lower performance metrics with slight variations in their results.

TABLE 7. Overall summary of models with different feature selection and parameter tuning methods.

Models	Feature Selection	Tuning	Accuracy	Precision	Recall	F1 Score	Specificity	ROC
LR	CSO	CSO	85.93	89.65	81.3	85.24	90.62	96.3
SVM	CSO	GA	86	90	81	85	91	92
XG + RF + CNN	GA+CSO	GA	88.6	86.96	83.3	85.11	91.67	94.49
GRU	CSO	GA	89	87.87	90.6	89.23	87.5	97.11
XG + RF + RNN	GA+CSO	GA	90	95	79.2	86.36	97.22	96.41
RF	CSO	CSO	90.6	90.6	90.5	90.6	90.6	97.11
RF+XG+NB	CSO	CSO	90.6	93.3	87.5	93.75	90.32	94.2
RF+ RNN+ BiLSTM	GA+CSO	GA	91.66	95.23	83.3	88.88	97.22	94.68
Random forest	GA+CSO	GA	91.67	91.3	87.5	89.36	94.44	96.53
XGBoost	GA+CSO	GA	91.67	95.24	83.3	88.89	97.22	94.91
CNN+BiLSTM+XGB	GA+CSO	GA	91.67	95.24	83.3	88.89	97.22	95.60
XGBoost	CSO	CSO	92.18	96.55	87.5	91.8	96.87	96.3
RF+CNN + weighted	GA+CSO	GA	93.3	95.45	87.5	91.3	97.22	94.10
RF +RNN+ weighted	GA+CSO	GA	95	95.65	91.7	93.61	97.22	94.06
RF [30]	-	Grid Search CV	82.6	87	85	85	-	0.95
MLP [31]	PSO	PSO	84.6	80.8	88.3	84.4	-	-
RF [32]	GA	Select K-Best	92	93.1	93.11	91.59	97.22	-
Proposed	GA+CSO	GA	95	95.65	91.7	93.61	97.22	95.02

Table 7 provide a thorough comparison of the proposed model with various existing ML models with different feature selection and parameter tuning methodologies.

It can be observed from Table 7 that in terms of accuracy, the best-performing models are the Proposed model and the RF+RNN+weighted ensemble, both with an accuracy of 95%. This is because the proposed model is benefitted using hybrid feature selection and ensemble approach, resulting in improved predictive performance. These are followed by the RF+CNN+weighted ensemble model, with an accuracy of 93.3%, and the XGBoost model, tuned with CSO, with an accuracy of 92.18%. In terms of precision, In terms of precision, XGBoost tuned with CSO is the top-performing model. This is followed by the proposed model and the RF+RNN+weighted ensemble, both with a score of 95.65%. The RF+CNN+Majority voting, RF+RNN+weighted ensemble, and RF+CNN+weighted ensemble models each have slight variations in their precision values. In terms of recall, the proposed model and the RF+CNN+weighted ensemble have the highest value of 91.7%, followed by the GRU and RF models, both tuned with CSO, with scores of 90.6% and 90.5%, respectively. The highest specificity of 97.22% is achieved by the proposed model, along with some other models.

Therefore, the proposed RF + CNN model with majority voting demonstrates superior performance across various metrics compared to traditional machine learning models and ensemble methods. The model achieved an accuracy of 95%, which is among the highest in the comparison table. It has a precision of 95.65%, indicating a high true positive rate. The recall score of 91.7% outperforms all other models, suggesting that our proposed model effectively identifies many positive cases. A higher recall value is crucial in disease prediction as it ensures that the maximum number of disease cases are detected correctly. The F1 score is 93.61%,

reflecting a good balance between recall and precision. With a specificity score of 97.22%, the proposed approach is highly effective at correctly identifying negative cases. Additionally, the ROC score of 95.02% confirms a good trade-off between the false positive and true positive rates. Overall, the proposed combination yields a model that is both precise and robust across various evaluation metrics.

VIII. THREATS TO VALIDITY

Although the UCI Heart Disease dataset is widely used and benchmarked, it consists of only 303 instances and may not fully capture the diversity and complexity present in large-scale clinical populations. While the proposed hybrid model demonstrates strong predictive performance on this dataset, its generalizability to more heterogeneous patient populations remain a critical consideration. To mitigate the risk of overfitting, our approach includes scalable design elements such as population-based feature selection, ensemble classification, and Genetic Algorithm-driven hyperparameter tuning. These elements enhance the model's robustness even on limited data. Nonetheless, future work will focus on validating the framework using larger and more diverse datasets, as well as exploring transfer learning or domain adaptation techniques to further improve generalization in real-world healthcare environments.

IX. CONCLUSION AND FUTURE WORK

This paper proposed an advanced method using a hybrid approach that combines the Cuckoo Search algorithm and the Genetic Algorithm for feature selection, followed by an ensemble of Convolutional Neural Networks and Random Forest using majority voting for prediction. The dataset used in this research is the UCI Heart Disease Dataset. The preprocessing techniques are removing outliers using the interquartile range (IQR) and standardization using

Min-Max normalization. The hybrid GA+CSA was used to select the most relevant features, and the hyperparameters were optimized using GA. Using the proposed approach, 95% accuracy, 95.65% precision, 91.7% recall, 93.61% F1-score, 97.22% specificity, and 95.02% ROC AUC has been achieved.

Future work is to validate the proposed approach on larger, more diverse clinical datasets to assess its scalability and generalization across real-world patient data. In addition, we will explore other optimization methods and include more sophisticated deep-learning models to improve the performance of the heart disease detection model.

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